

Quantum Tomography for High Energy Physics

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1 Quantum Tomography at work

1.1 One Spin-1/2 system

Let's describe the one spin-1/2 density matrix:

$$\rho = \frac{1}{2} (\mathcal{I} + \vec{B} \cdot \vec{\sigma}) \quad (1)$$

If we want to analyze a decay $\tau/\bar{\tau} \rightarrow l^+/l^-$, then the option of measure just two directions (up/down) isn't available, instead we need to measure in a space, in a solid angle space. So, we have the intention of measure in a specific direction in which the product of the decay goes on, say $\hat{n}$, then the projector operator will take the form:

$$\Pi_{\hat{n}} = \frac{1}{2} (\mathcal{I} + \hat{n} \cdot \vec{\sigma}) \quad (2)$$

The probability that the decay product be emitted in the $\hat{n}$ direction is:

$$\begin{aligned} P(\hat{n}) &= \text{Tr} \left(\frac{1}{2} [\mathcal{I} + \vec{B} \cdot \vec{\sigma}] \frac{1}{2} [\mathcal{I} + \hat{n} \cdot \vec{\sigma}] \right) \\ &= \text{Tr} \left(\frac{1}{4} [\mathcal{I} + \vec{B} \cdot \vec{\sigma} + \hat{n} \cdot \vec{\sigma} + (\vec{B} \cdot \vec{\sigma})(\hat{n} \cdot \vec{\sigma})] \right) \end{aligned} \quad (3)$$